

Gold Bot v2 — Moteur de Backtest Événementiel (Event-Driven)

Réplication fidèle de `bot_headless.py` : moteur hybride **M5 (signaux) / M1 (simulation intra-trade)**.

Architecture :

- Indicateurs calculés sur fenêtre glissante M5 (100 bougies)
- Détection de signal → score → filtres horaires
- Simulation des trades sur données M1 pour TP/SL précis
- Breakeven automatique, SL Cap 22 pts, Daily Stop -150 \$

Décalage horaire : CSV en GMT+3 (Broker) → soustraction de 1h pour passer en GMT+2 (PC/Bot).

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In [1]: # =====
# CELLULE 1 – Imports & Configuration globale
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import warnings
from datetime import timedelta
from pathlib import Path

warnings.filterwarnings('ignore')
plt.style.use('dark_background')

# — Chemins CSV (adapter si nécessaire) —————
PATH_M1 = Path('data/XAUUSD-VIP_M1_6Months.csv')
PATH_M5 = Path('data/XAUUSD-VIP_M5_6Months.csv')

# — Paramètres de trading (modifiables) —————
LOT_SIZE      = 0.03
PNL_PER_POINT = 3.0      # 0.03 lot × 100 $/point (XAU) = 3 $/point
MAX_RISK_PTS  = 22.0
DAILY_MAX_LOSS = -150.0
SESSION_MID_H = 12      # Heure PC (GMT+2)
SESSION_END_H = 15      # Heure PC (GMT+2)
GMT_OFFSET    = -1      # Décalage Broker→PC : GMT+3 → GMT+2
MAX_TRADES_PER_DAY = 60

# — Modes disponibles —————
MODES_CONFIG = {
    'safe':    {'sl_mult':2.5, 'tp1_mult':1.5, 'tp2_mult':2.5,
                'tp3_mult':4.0, 'score_min':60, 'cooldown':300},
    'risque':  {'sl_mult':1.8, 'tp1_mult':1.2, 'tp2_mult':2.0,
                'tp3_mult':3.2, 'score_min':45, 'cooldown':180},
    'risque+++': {'sl_mult':1.5, 'tp1_mult':1.0, 'tp2_mult':1.8,
                  'tp3_mult':2.8, 'score_min':35, 'cooldown':90},
}

# — Paramètre global de mode actif (modifiable pour tests) —
TRADING_MODE = 'risque'

print('✅ Imports et configuration OK')
print(f'   Mode : {TRADING_MODE} | Score min : {MODES_CONFIG[TRADING_MODE]["score_min"]} | SL mult : {MODES_CONFIG[TRADING_MODE]["sl_mult"]}')

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✅ Imports et configuration OK
   Mode : risque | Score min : 45 | SL mult : 1.8

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In [2]: # =====
# CELLULE 2 – Chargement & Alignement Temporel
# =====

def load_csv(path: Path, label: str) -> pd.DataFrame:
    """Charge un CSV OHLCV broker, applique le décalage GMT+3 →
    GMT+2."""
    df = pd.read_csv(path, parse_dates=['Time'])
    df.rename(columns={'TickVolume': 'Volume'}, inplace=True)
    # [FIX TIMEZONE] CSV en heure Broker (GMT+3) → PC Bot (GMT+2) : -1h
    df['Time'] = df['Time'] + pd.Timedelta(hours=GMT_OFFSET)
    df.set_index('Time', inplace=True)
    df.sort_index(inplace=True)
    df[['Open', 'High', 'Low', 'Close', 'Volume']] =
df[['Open', 'High', 'Low', 'Close', 'Volume']].astype(float)
    print(f'✅ {label} chargé : {len(df):,} bougies | {df.index[0]} →
{df.index[-1]}')
    return df

df_m5 = load_csv(PATH_M5, 'M5')
df_m1 = load_csv(PATH_M1, 'M1')

print(f'\nAperçu M5 (5 premières lignes):')
display(df_m5.head())
```

✅ M5 chargé : 34,822 bougies | 2025-11-10 00:00:00 → 2026-05-08 19:25:00

✅ M1 chargé : 173,813 bougies | 2025-11-10 00:00:00 → 2026-05-08 19:29:00

Aperçu M5 (5 premières lignes):

	Open	High	Low	Close	Volume
Time					
2025-11-10 00:00:00	4000.52	4010.63	3998.57	4006.41	1891.0
2025-11-10 00:05:00	4006.41	4010.38	4005.32	4009.36	1246.0
2025-11-10 00:10:00	4009.37	4011.58	4008.04	4010.71	1173.0
2025-11-10 00:15:00	4010.57	4011.51	4004.49	4006.44	1746.0
2025-11-10 00:20:00	4006.44	4010.86	4006.44	4010.51	1378.0

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In [3]: # =====
# CELLULE 3 – Fonctions Indicateurs (répliques exactes du bot)
# =====

def compute_vwap(df: pd.DataFrame, window: int = 50) -> pd.Series:
    """VWAP glissant – réplique exacte de bot_headless.compute_vwap."""
    typical = (df['High'] + df['Low'] + df['Close']) / 3
    tp_vol = typical * df['Volume']
    return tp_vol.rolling(window).sum() /
df['Volume'].rolling(window).sum()

def compute_volume_profile(df: pd.DataFrame, bins: int = 20) -> dict:
    """Volume Profile (POC + Value Area 70%) – réplique exacte."""
    price_min = df['Low'].min()
    price_max = df['High'].max()
    if price_max == price_min:
        return {'poc': df['Close'].iloc[-1],
                'value_area_high': price_max,
                'value_area_low': price_min}
    volumes, edges = np.histogram(df['Close'], bins=bins,
weights=df['Volume'])
    poc_idx = volumes.argmax()
    poc = (edges[poc_idx] + edges[poc_idx + 1]) / 2
    target = volumes.sum() * 0.70
    sorted_i = np.argsort(volumes)[::-1]
    acc, va_i = 0, []
    for idx in sorted_i:
        acc += volumes[idx]; va_i.append(idx)
        if acc >= target: break
    va_high = max((edges[i]+edges[i+1])/2 for i in va_i)
    va_low = min((edges[i]+edges[i+1])/2 for i in va_i)
    return {'poc': round(poc,2), 'value_area_high': round(va_high,2),
'value_area_low': round(va_low,2)}

def compute_atr(df: pd.DataFrame, window: int = 14) -> float:
    """ATR – réplique exacte."""
    tr = pd.concat([
        df['High'] - df['Low'],
        (df['High'] - df['Close'].shift()).abs(),
        (df['Low'] - df['Close'].shift()).abs(),
    ], axis=1).max(axis=1)
    return round(tr.rolling(window).mean().iloc[-1], 2)

def _detect_candle_pattern(df: pd.DataFrame) -> str:
    """Patterns bougies – réplique exacte."""
    o,h,l,c = df['Open'].iloc[-1], df['High'].iloc[-1],
df['Low'].iloc[-1], df['Close'].iloc[-1]
    body = abs(c-o); candle = h-l
    if candle == 0: return 'Doji'
    ratio = body/candle
    lower_wick = min(o,c)-l
    upper_wick = h-max(o,c)
    if ratio < 0.1:
return 'Doji'

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    if lower_wick > body*2 and upper_wick < body*0.5 and c > o:
return 'Marteau'
    if upper_wick > body*2 and lower_wick < body*0.5 and c < o:
return 'Shooting Star'
    prev_o,prev_c = df['Open'].iloc[-2], df['Close'].iloc[-2]
    if c>o and c>prev_o and o<prev_c and prev_c<prev_o:
return 'Engulfing haussier'
    if c<o and c<prev_o and o>prev_c and prev_c>prev_o:
return 'Engulfing baissier'
    if lower_wick > body*3:
return 'Pin Bar haussier'
    if upper_wick > body*3:
return 'Pin Bar baissier'
    return 'Neutre'

def compute_indicators(df: pd.DataFrame) -> dict:
    """Calcul complet des indicateurs – réplique exacte de
bot_headless.compute_indicators.
    Fenêtre Volume Profile : 100 dernières bougies (tail(100)).
    """
    close = df['Close']; high = df['High']; low = df['Low']; vol =
df['Volume']
    price = close.iloc[-1]
    atr = compute_atr(df)

    vwap_series = compute_vwap(df, window=50)
    vwap = vwap_series.iloc[-1]
    typical = (high+low+close)/3
    vwap_std = (typical - vwap_series).rolling(50).std().iloc[-1]
    if np.isnan(vwap_std) or vwap_std == 0: vwap_std = 1e-9

    vwap_upper1 = round(vwap + 1.0*vwap_std, 2)
    vwap_lower1 = round(vwap - 1.0*vwap_std, 2)
    vwap_upper2 = round(vwap + 2.0*vwap_std, 2)
    vwap_lower2 = round(vwap - 2.0*vwap_std, 2)
    dist_vwap = round(price - vwap, 2)
    dist_vwap_sd= round(dist_vwap / vwap_std, 2)
    vwap_slope = round(vwap_series.diff(5).iloc[-1], 2)

    vp = compute_volume_profile(df.tail(100))
    poc = vp['poc']
    near_poc= abs(price - poc) < atr * 0.5

    swing_high = round(df.tail(30)['High'].max(), 2)
    swing_low = round(df.tail(30)['Low'].min(), 2)
    near_swing_support = abs(price - swing_low) < atr * 1.0
    near_swing_resistance = abs(price - swing_high) < atr * 1.0

    ema200 = close.ewm(span=200, adjust=False).mean().iloc[-1]
    above_200 = price > ema200
    ema20 = close.ewm(span=20, adjust=False).mean()
    ema50 = close.ewm(span=50, adjust=False).mean()
    ema_cross_bull = (ema20.iloc[-1]>ema50.iloc[-1] and ema20.iloc[-2]
<=ema50.iloc[-2])
    ema_cross_bear = (ema20.iloc[-1]<ema50.iloc[-1] and
ema20.iloc[-2]>=ema50.iloc[-2])

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avg_vol    = vol.rolling(20).mean().iloc[-1]
vol_ratio  = round(vol.iloc[-1]/avg_vol, 2) if avg_vol>0 else 1.0
vol_spike  = vol_ratio > 1.4

cur_open   = df['Open'].iloc[-1]; cur_close = close.iloc[-1]
cur_body   = abs(cur_close-cur_open)
cur_bull   = cur_close > cur_open; cur_bear = cur_close < cur_open
last10     = close.iloc[-11:-1]
move10     = last10.iloc[-1]-last10.iloc[0]
above_50   = price > ema50.iloc[-1]

bounce_bull = (
    near_swing_support and cur_bull and cur_body > atr*0.5
    and vol_ratio > 1.4 and move10 < -atr*1.5
    and (not above_200 or (above_200 and near_poc))
)
bounce_bear = (
    near_swing_resistance and cur_bear and cur_body > atr*0.5
    and vol_ratio > 1.4 and move10 > atr*1.5
)
pattern = _detect_candle_pattern(df)

return {
    'price': round(price,2), 'atr': atr,
    'vwap': round(vwap,2), 'vwap_upper1': vwap_upper1,
'vwap_lower1': vwap_lower1,
    'vwap_upper2': vwap_upper2, 'vwap_lower2': vwap_lower2,
    'dist_vwap_sd': dist_vwap_sd, 'vwap_slope': vwap_slope,
    'poc': poc, 'va_high': vp['value_area_high'], 'va_low':
vp['value_area_low'],
    'near_poc': near_poc, 'swing_high': swing_high, 'swing_low':
swing_low,
    'near_swing_sup': near_swing_support, 'near_swing_res':
near_swing_resistance,
    'ema200': round(ema200,2), 'ema20': round(ema20.iloc[-1],2),
'ema50': round(ema50.iloc[-1],2),
    'above_200': above_200, 'above_50': above_50,
    'ema_cross_bull': ema_cross_bull, 'ema_cross_bear':
ema_cross_bear,
    'vol_ratio': vol_ratio, 'vol_spike': vol_spike,
    'bounce_bull': bounce_bull, 'bounce_bear': bounce_bear,
    'move10': round(move10,2), 'pattern': pattern,
}

print('✅ Fonctions indicateurs définies')

```

✅ Fonctions indicateurs définies

In [4]:

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# =====
# CELLULE 4 – Direction, Score & Niveaux TP/SL
# =====

def determine_direction(ind: dict):
    """Réplique exacte de bot_headless.determine_direction."""
    if abs(ind['vwap_slope']) < ind['atr'] * 0.1: # [FIX VWAP SLOPE]
        return None, 'vwap_flat'
    dist = ind['dist_vwap_sd']
    if dist > 4.5 or dist < -4.5: # [FIX SD CAP]
        return None, 'sd_cap'

    price = ind['price']
    above_200 = ind['above_200']
    above_50 = ind['above_50']

    if ind['bounce_bull']: return 'BUY', 'bounce_bull'
    if ind['bounce_bear']: return 'SELL', 'bounce_bear'

    vwap_long = price < ind['vwap_lower1']
    vwap_short = price > ind['vwap_upper1']
    struct_long = ind['near_swing_sup'] or ind['near_poc']
    struct_short = ind['near_swing_res'] or ind['near_poc']
    vol_ok = ind['vol_ratio'] > 1.0

    if vwap_long and struct_long and above_200 and above_50 and (vol_ok or ind['ema_cross_bull']):
        return 'BUY', 'vwap_1sd'
    if vwap_short and struct_short and not above_200 and not above_50 and (vol_ok or ind['ema_cross_bear']):
        return 'SELL', 'vwap_1sd'
    if price < ind['vwap_lower2'] and above_200 and vol_ok:
        return 'BUY', 'vwap_2sd'
    if price > ind['vwap_upper2'] and not above_200 and vol_ok:
        return 'SELL', 'vwap_2sd'
    return None, 'no_setup'

def compute_score(ind: dict, direction: str):
    """Réplique exacte de bot_headless.compute_score."""
    score = 0; reasons = []; is_buy = direction == 'BUY'
    dist_sd = abs(ind['dist_vwap_sd'])
    if dist_sd >= 2.0: score += 30; reasons.append(f'SD={dist_sd:.1f}>=2.0 (+30)')
    elif dist_sd >= 1.0: score += 18; reasons.append(f'SD={dist_sd:.1f}>=1.0 (+18)')
    elif dist_sd >= 0.5: score += 8; reasons.append(f'SD={dist_sd:.1f}>=0.5 (+8)')

    if is_buy and ind['vwap_slope'] > 0: score += 5; reasons.append('Slope+ (+5)')
    elif not is_buy and ind['vwap_slope'] < 0: score += 5; reasons.append('Slope- (+5)')

    if is_buy:
        if ind['near_swing_sup']: score += 15; reasons.append('SwingSup (+15)')

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        if ind['near_poc']:          score += 10; reasons.append('POC
(+10)')
        if ind['price'] < ind['va_low']:  score += 10;
reasons.append('BelowVAL (+10)')
        if ind['above_200']:          score += 5; reasons.append('Above200
(+5)')
    else:
        if ind['near_swing_res']: score += 15; reasons.append('SwingRes
(+15)')
        if ind['near_poc']:          score += 10; reasons.append('POC
(+10)')
        if ind['price'] > ind['va_high']: score += 10;
reasons.append('AboveVAH (+10)')
        if not ind['above_200']:  score += 5; reasons.append('Below200
(+5)')

    vr = ind['vol_ratio']
    if ind['bounce_bull'] or ind['bounce_bear']: score += 25;
reasons.append(f'Bounce (+25)')
    elif vr >= 2.0: score += 20; reasons.append(f'VolSpike x{vr} (+20)')
    elif vr >= 1.4: score += 12; reasons.append(f'VolHigh x{vr} (+12)')
    elif vr >= 1.0: score += 5; reasons.append(f'VolOK x{vr} (+5)')

    if is_buy and ind['ema_cross_bull']: score += 10;
reasons.append('GoldenX (+10)')
    elif not is_buy and ind['ema_cross_bear']: score += 10;
reasons.append('DeathX (+10)')

    bull_p = {'Marteau', 'Engulfing haussier', 'Pin Bar haussier'}
    bear_p = {'Shooting Star', 'Engulfing baissier', 'Pin Bar baissier'}
    if is_buy and ind['pattern'] in bull_p: score += 15;
reasons.append(f"{ind['pattern']} (+15)")
    elif not is_buy and ind['pattern'] in bear_p: score += 15;
reasons.append(f"{ind['pattern']} (+15)")
    elif ind['pattern'] == 'Doji': score += 3; reasons.append('Doji
(+3)')

    return min(score, 100), reasons

def calculate_tp_sl(direction: str, entry: float, atr: float,
                    swing_high: float, swing_low: float, mode: str =
TRADING_MODE) -> dict:
    """Réplique exacte de bot_headless.calculate_tp_sl avec SL Cap."""
    m = MODES_CONFIG[mode]
    if direction == 'BUY':
        sl_str = round(swing_low - atr*0.2, 2)
        sl_atr = round(entry - atr*m['sl_mult'], 2)
        sl = min(sl_str, sl_atr)
        sl_floor = round(entry - atr*max(m['sl_mult'], 1.5), 2)
        sl = min(sl, sl_floor)
        risk = abs(entry - sl)
        if risk > MAX_RISK_PTS:          # [FIX SL CAP]
            sl = round(entry - MAX_RISK_PTS, 2)
            risk = MAX_RISK_PTS
        return {'sl': sl,
                'tp1': round(entry + risk*m['tp1_mult'], 2),

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        'tp2': round(entry + risk*m['tp2_mult'],2),
        'tp3': round(entry + risk*m['tp3_mult'],2)}
else:
    sl_str = round(swing_high + atr*0.2, 2)
    sl_atr = round(entry + atr*m['sl_mult'], 2)
    sl      = max(sl_str, sl_atr)
    sl_ceil = round(entry + atr*max(m['sl_mult'],1.5), 2)
    sl      = max(sl, sl_ceil)
    risk    = abs(sl - entry)
    if risk > MAX_RISK_PTS:          # [FIX SL CAP]
        sl = round(entry + MAX_RISK_PTS, 2)
        risk = MAX_RISK_PTS
    return {'sl': sl,
            'tp1': round(entry - risk*m['tp1_mult'],2),
            'tp2': round(entry - risk*m['tp2_mult'],2),
            'tp3': round(entry - risk*m['tp3_mult'],2)}

print('✅ Fonctions signal & niveaux définies')

```

✅ Fonctions signal & niveaux définies

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In [5]: # =====
# CELLULE 5 (REMPLACEMENT) – Simulateur M1 optimisé
# =====
# Changements vs version originale :
# 1. df_m1.index.searchsorted() au lieu du masque booléen O(n)
# 2. bars['High'].values / bars['Low'].values au lieu d'iterrows
# Logique breakeven et priorité TP/SL : identique à l'original

def simulate_trade_m1(direction: str, entry: float, levels: dict,
                      open_time: pd.Timestamp, df_m1: pd.DataFrame) ->
list:
    """
    Simule le parcours de prix en M1 pour un trade ouvert à `open_time`.
    Version optimisée : searchsorted + tableaux numpy (pas d'iterrows).
    Logique métier identique à la version originale.
    """
    sl_orig = levels['sl']
    tp1, tp2, tp3 = levels['tp1'], levels['tp2'], levels['tp3']

    # — [OPT] Slicing rapide via searchsorted (O(log n) vs O(n)) —
    start_pos = df_m1.index.searchsorted(open_time)
    bars = df_m1.iloc[start_pos:]

    if bars.empty:
        return []

    positions = [
        {'id': 'P1', 'tp': tp1, 'sl': sl_orig,
         'result': None, 'exit_px': None, 'closed': False},
        {'id': 'P2', 'tp': tp2, 'sl': sl_orig,
         'result': None, 'exit_px': None, 'closed': False},
        {'id': 'P3', 'tp': tp3, 'sl': sl_orig,
         'result': None, 'exit_px': None, 'closed': False},
    ]
    tp1_hit = False
    tp2_hit = False

    # — [OPT] Extraire les tableaux numpy une seule fois —————
    highs = bars['High'].values
    lows = bars['Low'].values
    closes = bars['Close'].values

    for k in range(len(highs)):
        if all(p['closed'] for p in positions):
            break

        bar_high = highs[k]
        bar_low = lows[k]

        for p in positions:
            if p['closed']:
                continue

            sl_cur = p['sl']
            tp_cur = p['tp']

            if direction == 'BUY':

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        tp_hit = bar_high >= tp_cur
        sl_hit = bar_low <= sl_cur
    else:
        tp_hit = bar_low <= tp_cur
        sl_hit = bar_high >= sl_cur

    # Priorité : si les deux touchés, le plus proche de l'entrée
d'abord
    if tp_hit and sl_hit:
        dist_tp = abs(tp_cur - entry)
        dist_sl = abs(sl_cur - entry)
        tp_hit = dist_tp < dist_sl
        sl_hit = not tp_hit

    if tp_hit:
        p['closed'] = True
        p['exit_px'] = tp_cur
        pts = (tp_cur - entry) if direction == 'BUY' else (entry
- tp_cur)

        p['result'] = p['id'].replace('P', 'TP')
        p['pnl'] = round(pts * PNL_PER_POINT, 2)

    # Breakeven : TP1 touché → SL de P2 et P3 → entry
    if p['id'] == 'P1' and not tp1_hit:
        tp1_hit = True
        for other in positions:
            if not other['closed']:
                other['sl'] = entry          # [FIX
BREAKEVEN]

    # SL de P3 → TP1 après TP2
    if p['id'] == 'P2' and not tp2_hit and tp1_hit:
        tp2_hit = True
        for other in positions:
            if not other['closed']:
                other['sl'] = tp1

    elif sl_hit:
        p['closed'] = True
        p['exit_px'] = p['sl']
        pts = (entry - p['sl']) if direction == 'BUY' else
(p['sl'] - entry)
        if abs(p['sl'] - entry) < 0.01:
            p['result'] = 'BE'
            p['pnl'] = 0.0
        else:
            p['result'] = 'SL'
            p['pnl'] = round(-pts * PNL_PER_POINT, 2)

    # Positions encore ouvertes → fermées au dernier prix
    last_close = closes[-1]
    for p in positions:
        if not p['closed']:
            pts = (last_close - entry) if direction == 'BUY' else (entry
- last_close)
            p['result'] = 'OPEN_END'
            p['exit_px'] = last_close

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p['pnl']      = round(pts * PNL_PER_POINT, 2)
p['closed']   = True

return positions

print('✅ simulate_trade_m1 optimisé (searchsorted + numpy)')

✅ simulate_trade_m1 optimisé (searchsorted + numpy)
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In [6]: # =====
# CELLULE 6 – Moteur de Backtest Principal (Event-Driven)
# =====

def run_backtest(df_m5: pd.DataFrame,
                 df_m1: pd.DataFrame,
                 mode: str = TRADING_MODE,
                 score_min_override: float = None,
                 sl_mult_override: float = None) -> pd.DataFrame:
    """
    Moteur événementiel complet.
    Paramètres modifiables pour analyse de sensibilité :
        score_min_override : remplace le score_min du mode
        sl_mult_override   : remplace le sl_mult du mode
    """
    m = MODES_CONFIG[mode].copy()
    if score_min_override is not None: m['score_min'] =
score_min_override
    if sl_mult_override is not None: m['sl_mult'] = sl_mult_override

    trades_log      = [] # Journal complet de tous les trades
    signals_ignored = [] # Signaux filtrés

    # État journalier
    daily_pnl      = 0.0
    trades_today   = 0
    last_sl_time   = pd.Timestamp('1970-01-01')
    last_signal_ts = pd.Timestamp('1970-01-01')
    current_date   = None
    daily_stopped  = False

    # Compteurs pour l'analyse des filtres
    n_vwap_flat    = 0
    n_sd_cap       = 0
    n_no_setup     = 0
    n_score_low    = 0
    n_session_end  = 0
    n_daily_stop   = 0
    n_cooldown     = 0

    WINDOW = 100 # Fenêtre glissante pour les indicateurs

    candle_times = df_m5.index.tolist()

    for i in range(WINDOW, len(df_m5)):
        ts = candle_times[i]
        day = ts.date()

        # — Reset quotidien —————
        if day != current_date:
            current_date = day
            daily_pnl     = 0.0
            trades_today  = 0
            daily_stopped = False

        if daily_stopped:
            n_daily_stop += 1; continue

```

```

# — [FIX DAILY STOP] Coupe-circuit —————
if daily_pnl < DAILY_MAX_LOSS:
    daily_stopped = True
    n_daily_stop += 1; continue

if trades_today >= MAX_TRADES_PER_DAY:
    continue

# — [FIX CUTOFF] Filtre horaire —————
h = ts.hour
if h >= SESSION_END_H:
    n_session_end += 1; continue

# — Cooldown inter-signaux —————
if (ts - last_signal_ts).total_seconds() < m['cooldown']:
    n_cooldown += 1; continue

# — Calcul des indicateurs —————
window_df = df_m5.iloc[i-WINDOW:i+1].copy()

try:
    ind = compute_indicators(window_df)
except Exception:
    continue

# — Détermination de la direction —————
direction, reject_reason = determine_direction(ind)

if direction is None:
    if reject_reason == 'vwap_flat': n_vwap_flat += 1
    elif reject_reason == 'sd_cap': n_sd_cap += 1
    else: n_no_setup += 1
    signals_ignored.append({'ts':ts,'reason':reject_reason})
    continue

# — Score & Threshold —————
score, reasons = compute_score(ind, direction)

threshold = m['score_min']
if h >= SESSION_MID_H: # [FIX THRESHOLD
+10]
    threshold += 10
if (ts - last_sl_time).total_seconds() < 300: # [FIX POST-SL
+20]
    threshold += 20

if score < threshold:
    n_score_low += 1
    signals_ignored.append({'ts':ts,'reason':f'score={score}
<{threshold}'},
'direction':direction,'score':score,'threshold':threshold})
    continue

# — Calcul TP/SL —————
levels = calculate_tp_sl(direction, ind['price'], ind['atr'],

```

```

                                ind['swing_high'], ind['swing_low'],
mode=mode)
    # Appliquer sl_mult override si nécessaire
    if sl_mult_override is not None:
        levels = calculate_tp_sl(direction, ind['price'],
ind['atr'],
                                ind['swing_high'],
ind['swing_low'],
                                mode=mode)

    entry = ind['price']

    # — Simulation M1 —————
    positions = simulate_trade_m1(direction, entry, levels, ts,
df_m1)

    if not positions:
        continue

    seq_pnl = sum(p['pnl'] for p in positions)
    daily_pnl = round(daily_pnl + seq_pnl, 2)
    trades_today += 1
    last_signal_ts = ts

    # Pénalité post-SL : si une position se termine en SL
    if any(p['result'] == 'SL' for p in positions):
        last_sl_time = ts

    # Logging
    for p in positions:
        trades_log.append({
            'open_time': ts,
            'day': day,
            'direction': direction,
            'entry': entry,
            'atr': ind['atr'],
            'score': score,
            'threshold': threshold,
            'sl': levels['sl'],
            'tp1': levels['tp1'],
            'tp2': levels['tp2'],
            'tp3': levels['tp3'],
            'position': p['id'],
            'result': p['result'],
            'exit_px': p['exit_px'],
            'pnl': p['pnl'],
            'daily_pnl': daily_pnl,
            'hour': h,
            'vwap_slope': ind['vwap_slope'],
            'dist_sd': ind['dist_vwap_sd'],
            'pattern': ind['pattern'],
            'vol_ratio': ind['vol_ratio'],
        })

df_trades = pd.DataFrame(trades_log)
df_ignored = pd.DataFrame(signals_ignored)

```

```
filter_stats = {
    'vwap_flat':    n_vwap_flat,
    'sd_cap':       n_sd_cap,
    'no_setup':     n_no_setup,
    'score_low':    n_score_low,
    'session_end':  n_session_end,
    'daily_stop':   n_daily_stop,
    'cooldown':     n_cooldown,
}

return df_trades, df_ignored, filter_stats
```

```
# — Lancement du backtest —————
print('🚀 Lancement du backtest...')
df_trades, df_ignored, filter_stats = run_backtest(df_m5, df_m1,
mode=TRADING_MODE)
print(f'✅ Backtest terminé : {len(df_trades)} lignes de trades
({len(df_trades)//3 if len(df_trades)>0 else 0} séquences)')
```

```
🚀 Lancement du backtest...
✅ Backtest terminé : 108 lignes de trades (36 séquences)
```


In [7]:

```

# =====
# CELLULE 6b – Pré-calcul des indicateurs (à insérer après Cell 6)
# =====
# Principe : compute_indicators() est coûteux (rolling pandas sur 100
# bougies). Pour 40 combinaisons de paramètres, on évite de le
# recalculer
# 40x en le calculant UNE SEULE FOIS ici et en réutilisant le résultat.

import time
from collections import OrderedDict

def precompute_all_indicators(df_m5: pd.DataFrame, window: int = 100) ->
OrderedDict:
    """
    Pré-calcule et stocke les indicateurs pour chaque bougie M5.
    Appel unique (~même durée qu'un run_backtest normal).
    Résultat réutilisé par run_backtest_fast() sans recomputation.
    """
    t0 = time.time()
    candle_times = df_m5.index.tolist()
    precomp = OrderedDict()
    total = len(df_m5)

    for i in range(window, total):
        ts = candle_times[i]
        window_df = df_m5.iloc[i - window: i + 1]
        try:
            precomp[ts] = compute_indicators(window_df)
        except Exception:
            pass # Bougie ignorée (données manquantes)

    # Affichage de progression toutes les 5000 itérations
    if (i - window) % 5000 == 0:
        pct = (i - window) / (total - window) * 100
        elapsed = time.time() - t0
        eta = (elapsed / max(i - window, 1)) * (total - window -
(i - window))
        print(f' {pct:5.1f}% – {i - window:,}/{total - window:,}
bougies '
              f'| Écoulé : {elapsed:.0f}s | ETA : {eta:.0f}s')

    print(f'\n✅ Pré-calcul terminé : {len(precomp):,} timestamps en
{time.time()-t0:.1f}s')
    return precomp

def run_backtest_fast(precomp: OrderedDict,
                      df_m1: pd.DataFrame,
                      mode: str = TRADING_MODE,
                      score_min_override: float = None,
                      sl_mult_override: float = None) -> tuple:
    """
    Moteur de backtest optimisé pour la sensibilité.
    Réutilise les indicateurs pré-calculés (precomp) au lieu de les
    recalculer – logique de trading IDENTIQUE à run_backtest().

    Correction : sl_mult_override modifie temporairement MODES_CONFIG

```

```

pour que calculate_tp_sl() l'applique réellement (le code original
recalculait les levels sans appliquer l'override).
"""
m = MODES_CONFIG[mode].copy()
if score_min_override is not None:
    m['score_min'] = score_min_override

# Appliquer sl_mult temporairement dans MODES_CONFIG
# (calculate_tp_sl lit directement MODES_CONFIG[mode])
original_sl_mult = MODES_CONFIG[mode]['sl_mult']
if sl_mult_override is not None:
    MODES_CONFIG[mode]['sl_mult'] = sl_mult_override

trades_log      = []
signals_ignored = []

daily_pnl       = 0.0
trades_today    = 0
last_sl_time    = pd.Timestamp('1970-01-01')
last_signal_ts  = pd.Timestamp('1970-01-01')
current_date    = None
daily_stopped   = False

n_vwap_flat = n_sd_cap = n_no_setup = n_score_low = 0
n_session_end = n_daily_stop = n_cooldown = 0

try:
    for ts, ind in precomp.items():
        day = ts.date()

        # — Reset quotidien —————
        if day != current_date:
            current_date = day
            daily_pnl     = 0.0
            trades_today  = 0
            daily_stopped = False

        if daily_stopped:
            n_daily_stop += 1
            continue

        # — [FIX DAILY STOP] Coupe-circuit —————
        if daily_pnl < DAILY_MAX_LOSS:
            daily_stopped = True
            n_daily_stop += 1
            continue

        if trades_today >= MAX_TRADES_PER_DAY:
            continue

        # — [FIX CUTOFF] Filtre horaire —————
        h = ts.hour
        if h >= SESSION_END_H:
            n_session_end += 1
            continue

        # — Cooldown inter-signaux —————

```

```

if (ts - last_signal_ts).total_seconds() < m['cooldown']:
    n_cooldown += 1
    continue

# — Direction (utilise indicateurs pré-calculés) ———
direction, reject_reason = determine_direction(ind)
if direction is None:
    if reject_reason == 'vwap_flat': n_vwap_flat += 1
    elif reject_reason == 'sd_cap': n_sd_cap += 1
    else: n_no_setup += 1
    signals_ignored.append({'ts': ts, 'reason':
reject_reason})
    continue

# — Score & Threshold —————
score, _ = compute_score(ind, direction)
threshold = m['score_min']
if h >= SESSION_MID_H: #
[FIX THRESHOLD +10]
    threshold += 10
    if (ts - last_sl_time).total_seconds() < 300: #
[FIX POST-SL +20]
        threshold += 20

if score < threshold:
    n_score_low += 1
    signals_ignored.append({
        'ts': ts, 'reason': f'score={score}<{threshold}',
        'direction': direction, 'score': score, 'threshold':
threshold,
    })
    continue

# — Calcul TP/SL (MODES_CONFIG déjà mis à jour si override)
—
levels = calculate_tp_sl(
    direction, ind['price'], ind['atr'],
    ind['swing_high'], ind['swing_low'], mode=mode,
)

entry = ind['price']

# — Simulation M1 —————
positions = simulate_trade_m1(direction, entry, levels, ts,
df_m1)

if not positions:
    continue

seq_pnl = sum(p['pnl'] for p in positions)
daily_pnl = round(daily_pnl + seq_pnl, 2)
trades_today += 1
last_signal_ts = ts

if any(p['result'] == 'SL' for p in positions):
    last_sl_time = ts

for p in positions:

```

```

trades_log.append({
    'open_time': ts,
    'day': day,
    'direction': direction,
    'entry': entry,
    'atr': ind['atr'],
    'score': score,
    'threshold': threshold,
    'sl': levels['sl'],
    'tp1': levels['tp1'],
    'tp2': levels['tp2'],
    'tp3': levels['tp3'],
    'position': p['id'],
    'result': p['result'],
    'exit_px': p['exit_px'],
    'pnl': p['pnl'],
    'daily_pnl': daily_pnl,
    'hour': h,
    'vwap_slope': ind['vwap_slope'],
    'dist_sd': ind['dist_vwap_sd'],
    'pattern': ind['pattern'],
    'vol_ratio': ind['vol_ratio'],
})

finally:
    # Restaurer sl_mult original dans MODES_CONFIG dans tous les cas
    if sl_mult_override is not None:
        MODES_CONFIG[mode]['sl_mult'] = original_sl_mult

df_trades_out = pd.DataFrame(trades_log)
df_ignored_out = pd.DataFrame(signals_ignored)
filter_stats = {
    'vwap_flat': n_vwap_flat,
    'sd_cap': n_sd_cap,
    'no_setup': n_no_setup,
    'score_low': n_score_low,
    'session_end': n_session_end,
    'daily_stop': n_daily_stop,
    'cooldown': n_cooldown,
}

return df_trades_out, df_ignored_out, filter_stats

# — Lancer le pré-calcul (une seule fois, même durée qu'un run normal)
—
print('⌚ Pré-calcul des indicateurs en cours (à faire UNE SEULE FOIS)...')
precomp_indicators = precompute_all_indicators(df_m5)
print('✅ Indicateurs pré-calculés – prêt pour la sensibilité rapide.')

```

```

⌚ Pré-calcul des indicateurs en cours (à faire UNE SEULE FOIS)...
 0.0% – 0/34,722 bougies | Écoulé : 0s | ETA : 2390s
14.4% – 5,000/34,722 bougies | Écoulé : 25s | ETA : 148s
28.8% – 10,000/34,722 bougies | Écoulé : 49s | ETA : 121s
43.2% – 15,000/34,722 bougies | Écoulé : 75s | ETA : 98s
57.6% – 20,000/34,722 bougies | Écoulé : 99s | ETA : 73s
72.0% – 25,000/34,722 bougies | Écoulé : 123s | ETA : 48s

```

86.4% – 30,000/34,722 bougies | Écoulé : 149s | ETA : 23s

- ✓ Pré-calcul terminé : 34,722 timestamps en 171.6s
- ✓ Indicateurs pré-calculés – prêt pour la sensibilité rapide.

```

In [8]: # =====
# CELLULE 7 – Statistiques Détaillées
# =====

def compute_stats(df_trades: pd.DataFrame, filter_stats: dict) -> dict:
    if df_trades.empty:
        print('⚠️ Aucun trade dans le backtest.')
        return {}

    # — Séquences (groupées par open_time + direction) —————
    seq = df_trades.groupby(['open_time', 'direction'])
    ['pnl'].sum().reset_index()
    seq.columns = ['open_time', 'direction', 'seq_pnl']

    net_profit = round(df_trades['pnl'].sum(), 2)
    gross_win = df_trades[df_trades['pnl']>0]['pnl'].sum()
    gross_loss = abs(df_trades[df_trades['pnl']<0]['pnl'].sum())
    profit_factor= round(gross_win/gross_loss, 2) if gross_loss>0 else
float('inf')

    n_seq = len(seq)
    n_win_seq = (seq['seq_pnl']>0).sum()
    winrate_seq = round(n_win_seq/n_seq*100, 1) if n_seq>0 else 0

    # — Winrate par palier —————
    for tp in ['TP1', 'TP2', 'TP3', 'SL', 'BE']:
        n = (df_trades['result']==tp).sum()
        pn = df_trades[df_trades['result']==tp]['pnl'].sum()
        print(f' {tp:6s} : {n:4d} fois | PnL cumulé : {pn:+.2f} $')

    # — Drawdown maximum —————
    equity = df_trades.groupby('open_time')['pnl'].sum().cumsum()
    running_max = equity.cummax()
    drawdown = equity - running_max
    max_drawdown = round(drawdown.min(), 2)

    print('\n' + '='*55)
    print(f' MODE : {TRADING_MODE.upper()}')
    print(f' Période : {df_trades["open_time"].min().date()} → {df_trades["open_time"].max().date()}')
    print(f' Nb séquences : {n_seq}')
    print(f' Net Profit : {net_profit:+.2f} $')
    print(f' Profit Factor : {profit_factor}')
    print(f' Max Drawdown : {max_drawdown:.2f} $')
    print(f' Winrate (séq.) : {winrate_seq} %')
    print(f' Meilleure séq. : {seq["seq_pnl"].max():+.2f} $')
    print(f' Pire séq. : {seq["seq_pnl"].min():+.2f} $')
    print('-'*55)
    print(' Filtres :')
    for k,v in filter_stats.items():
        print(f' {k:<15} : {v:,}')
    print('='*55)

    return {
        'net_profit': net_profit, 'profit_factor': profit_factor,
        'max_drawdown': max_drawdown, 'winrate_seq': winrate_seq,
        'n_seq': n_seq, 'equity': equity,

```

```
}  
  
print('\n=== RÉSULTATS DU BACKTEST ===')  
print('Winrate par palier :')  
stats = compute_stats(df_trades, filter_stats)
```

=== RÉSULTATS DU BACKTEST ===

Winrate par palier :

TP1	:	24 fois		PnL cumulé	:	+852.72 \$
TP2	:	11 fois		PnL cumulé	:	+584.52 \$
TP3	:	1 fois		PnL cumulé	:	+60.30 \$
SL	:	46 fois		PnL cumulé	:	-818.22 \$
BE	:	26 fois		PnL cumulé	:	+0.00 \$

MODE	:	RISQUE
Période	:	2025-11-19 → 2026-05-06
Nb séquences	:	36
Net Profit	:	+679.32 \$
Profit Factor	:	1.59
Max Drawdown	:	-399.48 \$
Winrate (séq.)	:	66.7 %
Meilleure séq.	:	+290.40 \$
Pire séq.	:	-174.96 \$

Filtres :

vwap_flat	:	3,669
sd_cap	:	268
no_setup	:	18,587
score_low	:	1
session_end	:	11,802
daily_stop	:	359
cooldown	:	0

```

In [9]: # =====
# CELLULE 8 – Graphiques
# =====

if df_trades.empty:
    print('⚠️ Aucun trade à afficher.')
else:
    fig = plt.figure(figsize=(20, 14))
    fig.suptitle(f'Gold Bot v2 – Backtest {TRADING_MODE.upper()} | Mode Hybride M5/M1',
                 fontsize=15, fontweight='bold', color='white', y=0.98)

    # — (1) Equity Curve —————
    ax1 = fig.add_subplot(3, 2, (1, 2))
    equity = stats['equity']
    colors_eq = ['#00ff88' if v >= 0 else '#ff4466' for v in equity.values]
    ax1.fill_between(equity.index, equity.values, alpha=0.25,
                    color='#00ff88' if equity.iloc[-1]>=0 else '#ff4466')
    ax1.plot(equity.index, equity.values, linewidth=1.5, color='#00d4ff', label='Equity')
    ax1.axhline(0, color='white', linewidth=0.5, linestyle='--', alpha=0.4)
    ax1.set_title('📈 Courbe d\'Équité Cumulée', fontsize=12, color='#00d4ff')
    ax1.set_ylabel('PnL cumulé ($)', color='white')
    ax1.tick_params(colors='white'); ax1.grid(alpha=0.15)
    # Annotation final
    ax1.annotate(f"Net: {equity.iloc[-1]:+.0f}$",
                xy=(equity.index[-1], equity.iloc[-1]),
                xytext=(-60, 15), textcoords='offset points',
                color='#ffdd44', fontsize=11, fontweight='bold',
                arrowprops=dict(arrowstyle='->', color='#ffdd44'))

    # — (2) PnL journalier en barres —————
    ax2 = fig.add_subplot(3, 2, 3)
    daily = df_trades.groupby('day')['pnl'].sum()
    bar_colors = ['#00ff88' if v>=0 else '#ff4466' for v in daily.values]
    ax2.bar(range(len(daily)), daily.values, color=bar_colors, alpha=0.85)
    ax2.axhline(0, color='white', linewidth=0.5)
    ax2.axhline(DAILY_MAX_LOSS, color='#ff8800', linewidth=1.2, linestyle='--', label=f'Daily Stop {DAILY_MAX_LOSS}$')
    ax2.set_title('📊 PnL Journalier', fontsize=11, color='#00d4ff')
    ax2.set_ylabel('PnL ($)', color='white')
    ax2.tick_params(colors='white'); ax2.grid(alpha=0.15, axis='y')
    ax2.legend(fontsize=8)

    # — (3) Répartition des résultats —————
    ax3 = fig.add_subplot(3, 2, 4)
    result_counts = df_trades['result'].value_counts()
    palette = {'TP1': '#00ff88', 'TP2': '#00cc66', 'TP3': '#009944',
               'SL': '#ff4466', 'BE': '#ffdd44', 'OPEN_END': '#888888'}
    colors_pie = [palette.get(r, '#aaaaaa') for r in result_counts.index]
    wedges, texts, autotexts = ax3.pie(

```



```

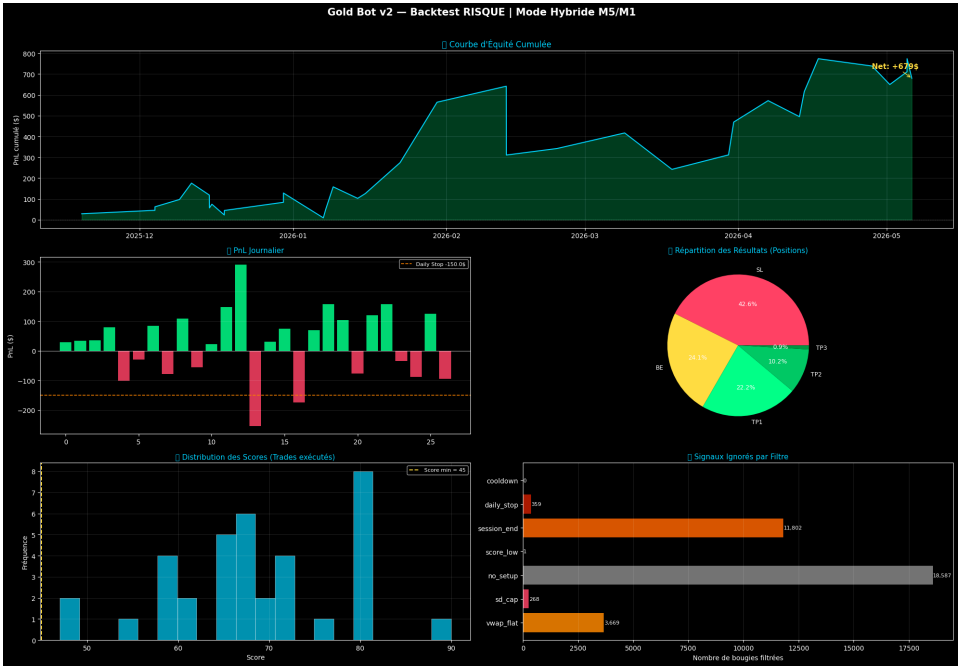
        result_counts.values, labels=result_counts.index,
        autopct='%1.1f%%', colors=colors_pie,
        textprops={'color':'white','fontsize':9})
    ax3.set_title('📊 Répartition des Résultats (Positions)',
    fontsize=11, color='#00d4ff')

    # — (4) Distribution des scores —————
    ax4 = fig.add_subplot(3, 2, 5)
    scores_traded = df_trades.groupby('open_time')['score'].first()
    ax4.hist(scores_traded.values, bins=20, color='#00d4ff', alpha=0.7,
    edgecolor='white', linewidth=0.5)
    ax4.axvline(MODES_CONFIG[TRADING_MODE]['score_min'],
    color='#ffdd44',
                linewidth=1.5, linestyle='--', label=f'Score min =
    {MODES_CONFIG[TRADING_MODE]["score_min"]}')
    ax4.set_title('📊 Distribution des Scores (Trades exécutés)',
    fontsize=11, color='#00d4ff')
    ax4.set_xlabel('Score', color='white')
    ax4.set_ylabel('Fréquence', color='white')
    ax4.tick_params(colors='white'); ax4.grid(alpha=0.15)
    ax4.legend(fontsize=8)

    # — (5) Impact des filtres —————
    ax5 = fig.add_subplot(3, 2, 6)
    labels_f = list(filter_stats.keys())
    values_f = list(filter_stats.values())
    colors_f =
    ['#ff8800','#ff4466','#888888','#dd4400','#ff6600','#cc2200','#996600']
    bars = ax5.barh(labels_f, values_f, color=colors_f[:len(labels_f)],
    alpha=0.85)
    ax5.set_title('🚦 Signaux Ignorés par Filtre', fontsize=11,
    color='#00d4ff')
    ax5.set_xlabel('Nombre de bougies filtrées', color='white')
    ax5.tick_params(colors='white'); ax5.grid(alpha=0.15, axis='x')
    for bar, val in zip(bars, values_f):
        ax5.text(bar.get_width()+10, bar.get_y()+bar.get_height()/2,
            f'{val:,}', va='center', color='white', fontsize=8)

    plt.tight_layout(rect=[0, 0, 1, 0.97])
    plt.savefig('backtest_results.png', dpi=150, bbox_inches='tight',
                facecolor='#d0d0d0', edgecolor='none')
    plt.show()
    print('✅ Graphique sauvegardé → backtest_results.png')

```



✓ Graphique sauvegardé → backtest_results.png

In [10]:

```

# =====
# CELLULE 9 – Graphique de Prix avec Trades (échantillonnage)
# =====

def plot_trades_on_price(df_m5: pd.DataFrame, df_trades: pd.DataFrame,
                        n_days: int = 3):
    """Affiche les trades sur le graphique de prix pour les N premiers
    jours tradés."""
    if df_trades.empty:
        print('⚠️ Aucun trade à afficher.');
```

return

```

    traded_days = sorted(df_trades['day'].unique())[:n_days]
    if not traded_days:
        print('⚠️ Aucun jour tradé.');
```

return

```

    fig, axes = plt.subplots(n_days, 1, figsize=(20, 5*n_days))
    if n_days == 1: axes = [axes]

    for ax, day in zip(axes, traded_days):
        # Données M5 du jour
        day_m5 = df_m5[df_m5.index.date == day]
        if day_m5.empty: continue

        xs = range(len(day_m5))
        ax.fill_between(xs,
                        day_m5['Low'].values - 0.5,
                        day_m5['High'].values + 0.5,
                        alpha=0.12, color='gray')
        ax.plot(xs, day_m5['Close'].values, color='#aaaaaa',
                linewidth=0.8, label='Close M5')

        # Trades du jour
        day_trades = df_trades[df_trades['day'] == day]

        for _, seq in day_trades.groupby('open_time'):
            row = seq.iloc[0]
            # Position sur l'axe X
            try:
                xi = day_m5.index.get_loc(row['open_time'],
method='nearest')
            except Exception:
                continue

            color = '#00ff88' if row['direction']=='BUY' else '#ff4466'
            marker = '^' if row['direction']=='BUY' else 'v'
            ax.scatter(xi, row['entry'], marker=marker, color=color,
s=100, zorder=5)

            # Niveaux
            for lvl, lbl, lc in [
                (row['s1'], 'SL', '#ff4466'),
                (row['tp1'], 'TP1', '#00ff88'),
                (row['tp2'], 'TP2', '#00cc66'),
                (row['tp3'], 'TP3', '#009944'),
            ]:
                ax.axhline(lvl, color=lc, linewidth=0.6, linestyle=':',
alpha=0.5)

```

```

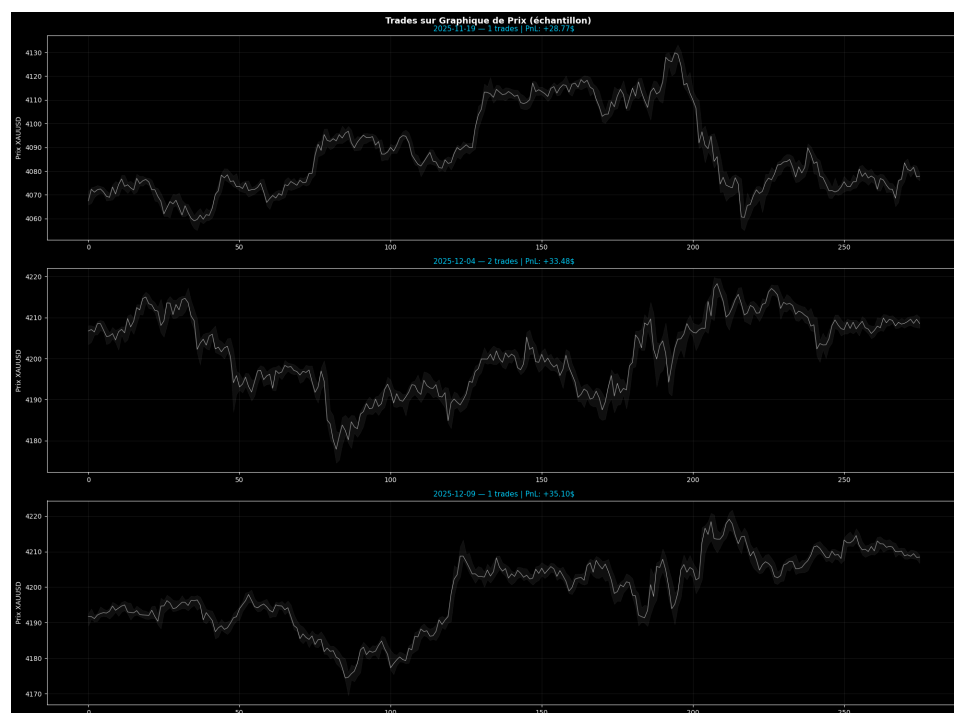
# PnL total
total_pnl = seq['pnl'].sum()
ax.annotate(f'{total_pnl:+.1f}$',
            (xi, row['entry']),
            textcoords='offset points', xytext=(5, 8 if
row['direction']=='BUY' else -14),
            color='#ffdd44', fontsize=7, fontweight='bold')

ax.set_title(f'{day} - {len(day_trades)//3} trades | PnL:
{day_trades["pnl"].sum():+.2f}$',
            fontsize=11, color='#00d4ff')
ax.set_ylabel('Prix XAUUSD', color='white')
ax.tick_params(colors='white')
ax.grid(alpha=0.10)

fig.suptitle('Trades sur Graphique de Prix (échantillon)',
            fontsize=13,
            fontweight='bold', color='white')
plt.tight_layout()
plt.savefig('backtest_price_chart.png', dpi=130,
            bbox_inches='tight',
            facecolor='#0d0d0d', edgecolor='none')
plt.show()
print('✅ Graphique prix sauvegardé → backtest_price_chart.png')

plot_trades_on_price(df_m5, df_trades, n_days=3)

```



✅ Graphique prix sauvegardé → backtest_price_chart.png

```

In [11]: # =====
# CELLULE 10 (REMPLACEMENT) – Analyse de Sensibilité RAPIDE
# =====
# Utilise precomp_indicators + run_backtest_fast.
# Les 40 combinaisons ne recalculent plus les indicateurs.

print('📊 Analyse de sensibilité rapide en cours...\n')
t_sens = time.time()

score_range = [35, 40, 45, 50, 55, 60, 65, 70]
sl_range    = [1.2, 1.5, 1.8, 2.2, 2.5]

results_matrix = []

for sm in score_range:
    for slm in sl_range:
        try:
            dt, _, fs = run_backtest_fast(
                precomp_indicators, df_m1,
                mode=TRADING_MODE,
                score_min_override=sm,
                sl_mult_override=slm,
            )
            if dt.empty:
                results_matrix.append({
                    'score_min': sm, 'sl_mult': slm,
                    'net_profit': 0, 'n_seq': 0, 'winrate': 0, 'pf': 0,
                })
                continue

            seq = dt.groupby('open_time')['pnl'].sum()
            net = round(seq.sum(), 2)
            n_seq = len(seq)
            wr = round((seq > 0).sum() / n_seq * 100, 1) if n_seq > 0
        else 0

            g1 = abs(dt[dt['pnl'] < 0]['pnl'].sum())
            gw = dt[dt['pnl'] > 0]['pnl'].sum()
            pf = round(gw / g1, 2) if g1 > 0 else 999

            results_matrix.append({
                'score_min': sm, 'sl_mult': slm,
                'net_profit': net, 'n_seq': n_seq, 'winrate': wr, 'pf':
pf,
            })

        except Exception as e:
            results_matrix.append({
                'score_min': sm, 'sl_mult': slm,
                'net_profit': 0, 'n_seq': 0, 'winrate': 0, 'pf': 0,
            })

print(f'✅ Sensibilité terminée en {time.time()-t_sens:.1f}s '
      f'({len(score_range)*len(sl_range)} combinaisons)')

df_sens = pd.DataFrame(results_matrix)

# — Heatmap Net Profit & Winrate (identique à l'original) —

```

```

pivot    = df_sens.pivot(index='sl_mult', columns='score_min',
values='net_profit')
pivot_wr = df_sens.pivot(index='sl_mult', columns='score_min',
values='winrate')

fig, axes = plt.subplots(1, 2, figsize=(18, 6))
fig.suptitle('Analyse de Sensibilité – Net Profit & Winrate',
             fontsize=13, fontweight='bold', color='white')

im1 = axes[0].imshow(pivot.values, aspect='auto', cmap='RdYlGn')
axes[0].set_xticks(range(len(pivot.columns)))
axes[0].set_xticklabels(pivot.columns, color='white')
axes[0].set_yticks(range(len(pivot.index)))
axes[0].set_yticklabels(pivot.index, color='white')
axes[0].set_xlabel('Score Min', color='white')
axes[0].set_ylabel('SL Mult', color='white')
axes[0].set_title('Net Profit ($)', color='#00d4ff')
for i in range(len(pivot.index)):
    for j in range(len(pivot.columns)):
        axes[0].text(j, i, f"{pivot.values[i,j]:.0f}",
                     ha='center', va='center', fontsize=8,
                     color='black' if abs(pivot.values[i,j]) < 200 else
                     'white')
plt.colorbar(im1, ax=axes[0])

im2 = axes[1].imshow(pivot_wr.values, aspect='auto', cmap='RdYlGn',
vmin=30, vmax=80)
axes[1].set_xticks(range(len(pivot_wr.columns)))
axes[1].set_xticklabels(pivot_wr.columns, color='white')
axes[1].set_yticks(range(len(pivot_wr.index)))
axes[1].set_yticklabels(pivot_wr.index, color='white')
axes[1].set_xlabel('Score Min', color='white')
axes[1].set_title('Winrate % (séquences)', color='#00d4ff')
for i in range(len(pivot_wr.index)):
    for j in range(len(pivot_wr.columns)):
        axes[1].text(j, i, f"{pivot_wr.values[i,j]:.0f}%",
                     ha='center', va='center', fontsize=8,
                     color='black')
plt.colorbar(im2, ax=axes[1])

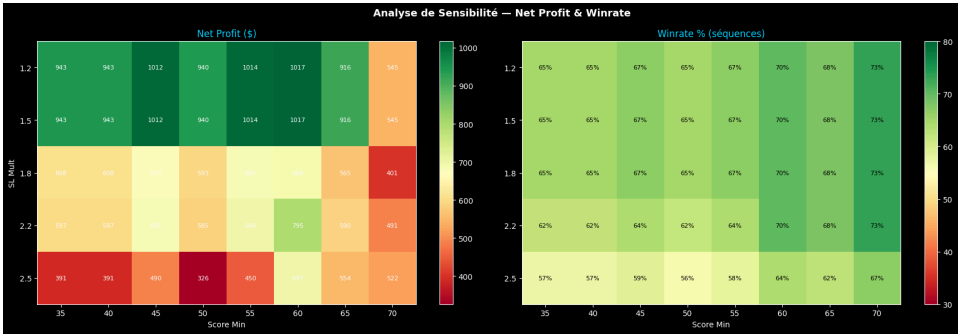
plt.tight_layout()
plt.savefig('sensitivity_analysis.png', dpi=130, bbox_inches='tight',
          facecolor='#0d0d0d', edgecolor='none')
plt.show()
print('✅ Analyse de sensibilité terminée → sensitivity_analysis.png')

# Meilleure combinaison
best = df_sens.loc[df_sens['net_profit'].idxmax()]
print(f'\n🏆 Meilleure combinaison : score_min={best["score_min"]} | '
      f'sl_mult={best["sl_mult"]} → Net: {best["net_profit"]:+.2f}$ | '
      f'WR: {best["winrate"]}%' )

```

 Analyse de sensibilité rapide en cours...

 Sensibilité terminée en 8.3s (40 combinaisons)



✓ Analyse de sensibilité terminée → sensitivity_analysis.png

🏆 Meilleure combinaison : score_min=60.0 | sl_mult=1.2 → Net: +1016.97\$ | WR: 70.4%

In [12]:

```

# =====
# CELLULE 11 – Analyse Détaillée des Filtrés & Statistiques Avancées
# =====

if not df_trades.empty:
    print('=== ANALYSE DES FILTRES HORAIRES ===')
    print('\nSignaux ignorés par filtre :')
    total_ignored = sum(filter_stats.values())
    for k, v in filter_stats.items():
        pct = v/total_ignored*100 if total_ignored>0 else 0
        bar = '█' * int(pct/2)
        print(f' {k:<15} : {v:6,} ({pct:4.1f}%) {bar}')

    print(f'\n Total ignorés : {total_ignored:,}')
    print(f' Total exécutés: {len(df_trades)//3}')

    print('\n=== STATS PAR HEURE D\'OUVERTURE ===')
    hourly = df_trades.groupby('hour').agg(
        n_trades=('pnl', 'count'),
        total_pnl=('pnl', 'sum'),
        avg_pnl=('pnl', 'mean'),
    ).round(2)
    hourly['n_seq'] = hourly['n_trades']//3
    display(hourly)

    print('\n=== STATS PAR DIRECTION ===')
    dir_stats = df_trades.groupby('direction').agg(
        n_pos=('pnl', 'count'),
        total_pnl=('pnl', 'sum'),
        win_rate=('pnl', lambda x: (x>0).mean()*100),
    ).round(2)
    display(dir_stats)

    print('\n=== TOP 5 MEILLEURES SÉQUENCES ===')
    seq_pnl = df_trades.groupby(['open_time', 'direction']).agg(
        total_pnl=('pnl', 'sum'), entry=('entry', 'first'), score=
('score', 'first')
    ).reset_index().sort_values('total_pnl', ascending=False)
    display(seq_pnl.head())

    print('\n=== TOP 5 PIRES SÉQUENCES ===')
    display(seq_pnl.tail())

    print('\n=== IMPACT SESSION_MID (12h) ===')
    before_mid = df_trades[df_trades['hour'] < SESSION_MID_H]
    after_mid = df_trades[df_trades['hour'] >= SESSION_MID_H]
    print(f' Avant 12h : {len(before_mid)//3} séq. | PnL:
{before_mid["pnl"].sum():+.2f}$')
    print(f' Après 12h : {len(after_mid)//3} séq. | PnL:
{after_mid["pnl"].sum():+.2f}$')

else:
    print('⚠️ Aucun trade dans le backtest – vérifiez vos données
CSV.')

```


Signaux ignorés par filtre :

Total ignorés : 34,686
Total exécutés: 36

	n_trades	total_pnl	avg_pnl	n_seq
hour				
0	18	286.62	15.92	6
1	21	28.38	1.35	7
2	33	58.14	1.76	11
3	3	157.74	52.58	1
6	3	28.77	9.59	1
7	3	120.60	40.20	1
8	6	-45.66	-7.61	2
11	3	78.66	26.22	1
13	3	-174.96	-58.32	1
14	15	141.03	9.40	5

	n_pos	total_pnl	win_rate
direction			
BUY	48	915.66	52.08
SELL	60	-236.34	35.00

	open_time	direction	total_pnl	entry	score
18	2026-01-30 02:10:00	BUY	290.40	5309.99	67

	open_time	direction	total_pnl	entry	score
30	2026-04-17 03:30:00	BUY	157.74	4774.17	80
26	2026-03-31 01:10:00	BUY	157.50	4498.78	58
17	2026-01-22 14:45:00	BUY	148.26	4815.24	80
29	2026-04-14 07:35:00	BUY	120.60	4759.13	72

=== TOP 5 PIRES SÉQUENCES ===

	open_time	direction	total_pnl	entry	score
35	2026-05-06 01:15:00	SELL	-94.41	4585.77	80
12	2026-01-07 02:35:00	BUY	-119.25	4470.77	68
21	2026-02-13 02:25:00	SELL	-163.80	4969.88	55
20	2026-02-13 02:20:00	SELL	-166.86	4972.26	68
24	2026-03-18 13:30:00	BUY	-174.96	4956.39	80

=== IMPACT SESSION_MID (12h) ===
Avant 12h : 30 séq. | PnL: +713.25\$
Après 12h : 6 séq. | PnL: -33.93\$

Notes d'utilisation

Changer de mode

```
`python
TRADING_MODE = 'safe' # 'safe' | 'risque' | 'risque+++'
`
```

Tester un score_min personnalisé

```
`python
df_trades, _ fs = run_backtest(df_m5, df_m1, mode='risque', score_min_override=55)
stats = compute_stats(df_trades, fs)
`
```

Tester un sl_mult personnalisé

```
`python
df_trades, _ fs = run_backtest(df_m5, df_m1, mode='risque', sl_mult_override=2.0)
`
```

Constantes reproductibles du bot

Paramètre	Valeur	Rôle
MAX_RISK_PTS	22.0	SL Cap absolu
DAILY_MAX_LOSS	-150 \$	Coupe-circuit journalier
SESSION_END_H	15	Arrêt des nouveaux trades
SESSION_MID_H	12	Threshold +10
GMT_OFFSET	-1	Décalage broker GMT+3 → PC GMT+2
PNL_PER_POINT	3.0 \$	0.03 lot × 100\$/pt
LOT_SIZE	0.03	Taille de lot par position

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